

Codor

INTELLIGENT PROGRAMMING EDUCATION PLATFORM

Hagenberger Fünfeck @ GDG

Track C: Next-Gen Education

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Programming Exercises in Education

01

EXERCISE CREATION

Creating quality programming exercises requires significant time investment for teachers.

02

INDIVIDUAL SUPPORT

Supporting students 1 on 1 at home is basically impossible.

03

VISIBILITY GAP

What are students struggling with at home?

04

ASSESSMENT SCALE

Grading at scale takes a lot of time.

Introducing Codor

An intelligent platform that transforms how programming exercises are created, assigned, and completed

FOR EDUCATORS

Upload solution code and generate structured exercises automatically

FOR STUDENTS

Receive contextual guidance without direct solutions

Creating an Assignment

←

<>

Teacher Dashboard

Create a new coding task

↑

Publish Task

🔄

Exercise Title

Build a Greeting Function

Teacher's Solution Code

```
1 function greet(name) {
2   const greeting = "Hello, " + name + "!";
3   console.log(greeting);
4   return greeting;
5 }
6
7 const result = greet("Student");
8 console.log(result);
```

🎯 Task Description

AI-generated task overview (editable)

In this exercise, you'll learn how to create a JavaScript function that takes a parameter and returns a personalized greeting. This is a fundamental skill in programming that teaches you about functions, parameters, string manipulation, and return values. You'll also practice logging output to the console for debugging purposes.

📋 Learning Steps

AI-generated steps linked to code lines

1 Lines 1-1

Define the function with the name parameter

2 Lines 2-2

Create a greeting variable using string concatenation

3 Lines 3-3

Log the greeting message to the console

Intelligent Guidance System

<>

CodeMentor

Interactive Programming Tutor

Lesson 1: Functions

🔄

lesson-01.js

Copy

Run

```
1 function greet(name) {
2   // This function greets the user
3   const message = "Hello, " + name + "!";
4   //?
5   return message;
6 }
7
8 // Call the function
9 const result = greet("Learner");
10 //?
```

OUTPUT

Run code to see output...

📖

Exercise

In this exercise, you'll learn how to create and use functions in JavaScript. Your goal is to:

- Create a function called 'greet' that takes a person's name as input
- The function should create a greeting message with that name
- Print the greeting to the console
- Return the greeting so it can be used elsewhere
- Call the function with a name and use the returned value...

⌵ Show more

🔔 Need Guidance?

Stuck on something? I'll analyze your code and provide helpful hints!

Get Help

I am not sure about the api call anymore

👤

🗨

Great question! It can be a little tricky to remember all the different ways to display information in JavaScript.

Instead of an "API call," in JavaScript, we often use built-in functions to interact with the environment. When you want to display something in the console, the most common function to use is `console.log()`.

Ask your tutor a question...

📤

Press Enter to send • Shift+Enter for new line

Analytics Dashboard

<

>

Student Progress

Track and analyze student performance

EXERCISES

Build a Greeting Function

3 students

Completed

2/3

Array Manipulation

2 students

Completed

1/2

STUDENTS

Alex Chen

completed

25 min

12 AI chats

95%

Maria Garcia

completed

32 min

18 AI chats

88%

James Wilson

in progress

15 min

8 AI chats

Maria Garcia

Build a Greeting Function

Time Spent

32 min

AI Interactions

18

Score

88%

AI Tutor Interactions

Questions and topics discussed with the AI tutor

1

Struggled with console.log placement

2

Asked about the difference between return and console.log

3

Needed clarification on variable scope

Score Deductions

Reasons why points were deducted from the final score

1

Incorrect variable naming convention

-4 pts

2

Missing error handling

-5 pts

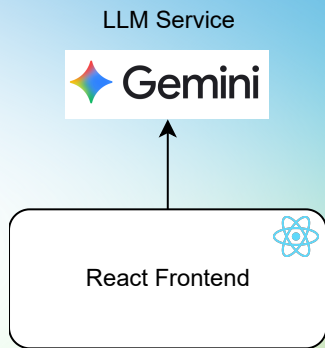
3

Code not properly indented

-3 pts

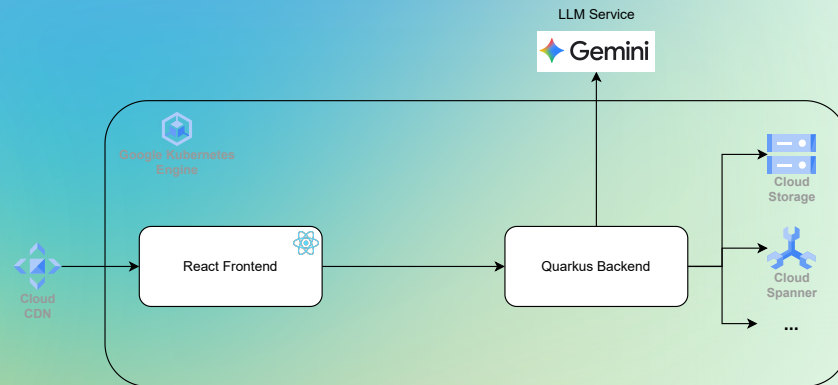
Architecture

CURRENT POC



Simple stack, everything is in the frontend, POC life 🙌

PRODUCTION SYSTEM



Scalable Architecture:

- Microservices backend
- Kubernetes deployment
- Cloud CDN for frontend
- Sandboxed code execution
- SSO integration
- ...

Codor

Enhancing teaching effectiveness
Enabling independent learning

Transforming Programming Education

GDG DevFest Submission 2025